1

(1) 设 $\varphi(t)=(a\cos t,a\sin t)$，于是

$$\varphi^*\omega=\frac{a\cos t\,\mathrm{d}(a\sin t)-a\sin t\,\mathrm{d}(a\cos t)}{a^2}=\frac{a^2\cos^2 t+a^2\sin^2 t}{a^2}\,\mathrm{d}t=\mathrm{d}t$$

于是

$$\int_\Gamma\frac{x\,\mathrm{d}y-y\,\mathrm{d}x}{x^2+y^2}=\int_0^{2\pi}\mathrm{d}t=2\pi$$

(2) 设 $\varphi(t)=(a\cos t,b\sin t)$。于是

$$\varphi^*\omega=(a\cos t+b\sin t)\,\mathrm{d}(a\cos t)+(a\cos t-b\sin t)\,\mathrm{d}(b\sin t)=(ab(\cos^2 t-\sin^2 t)-(a^2+b^2)\sin t\cos t)\,\mathrm{d}t$$

于是

$$\begin{aligned}\int_\Gamma(x+y)\,\mathrm{d}x+(x-y)\,\mathrm{d}y&=\int_0^{2\pi}(ab(\cos^2 t-\sin^2 t)-(a^2+b^2)\sin t\cos t)\,\mathrm{d}t\\&=ab\int_0^{2\pi}\cos(2t)\,\mathrm{d}t-\frac{a^2+b^2}{2}\int_0^{2\pi}\sin(2t)\,\mathrm{d}t=0\end{aligned}$$

(3) 设 $\varphi(y)=(y^2,y)$，于是

$$\varphi^*\omega=(y^4-2y^3)\,\mathrm{d}(y^2)+(y^2-2y^3)\,\mathrm{d}y=(2y^5-4y^4-2y^3+y^2)\,\mathrm{d}y$$

$$\int_\Gamma(x^2-2xy)\,\mathrm{d}x+(y^2-2xy)\,\mathrm{d}y=\int_{-1}^1(2y^5-4y^4-2y^3+y^2)\,\mathrm{d}y=-\frac{14}{15}$$

(4) 分成三个线段计算 $\Gamma_1:(0,0)\to(\frac12,0),\Gamma_2:(\frac12,0)\to(0,1),\Gamma_3:(0,1)\to(0,0)$:

$$\int_{\Gamma_1}x\,\mathrm{d}y=0,\int_{\Gamma_2}x\,\mathrm{d}y=\int_0^1\frac{1-y}{2}\,\mathrm{d}y=\frac14,\int_{\Gamma_3}x\,\mathrm{d}y=0$$

于是

$$\int_\Gamma x\,\mathrm{d}y=\int_{\Gamma_1}x\,\mathrm{d}y+\int_{\Gamma_2}x\,\mathrm{d}y+\int_{\Gamma_3}x\,\mathrm{d}y=\frac14$$

(5) 同样的分段计算

$$\int_\Gamma(x^2+y^2)\,\mathrm{d}y=\int_4^1(1^2+y^2)\,\mathrm{d}y+\int_1^4(3^2+y^3)\,\mathrm{d}y=24$$

2

设 $\varphi(t)=(\cos t,\sin t)$，于是

$$\varphi^*\omega=\frac{\mathrm{d}t}{a\cos^2 t+2b\sin t\cos t+c\sin^2 t}$$

$$
\begin{aligned}
\text{原式} &= \int_0^{2\pi} \frac{\mathrm{d}t}{a\cos^2 t + 2b\sin t\cos t + c\sin^2 t} = 2\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\mathrm{d}t}{a\cos^2 t + 2b\sin t\cos t + c\sin^2 t} \\
&= 2\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos^{-2} t}{a\tan^2 t + 2b\tan t + c}\,\mathrm{d}t \xlongequal{u=\tan t} 2\int_{-\infty}^{+\infty} \frac{\mathrm{d}u}{au^2+2bu+c}
\end{aligned}
$$

其中

$$
\int \frac{\mathrm{d}u}{au^2+2bu+c} = \frac{1}{a}\int \frac{\mathrm{d}u}{(u+\frac{b}{a})^2 + \frac{ac-b^2}{a^2}} = \frac{1}{a}\frac{a}{\sqrt{ac-b^2}}\arctan\frac{au+b}{ac-b^2} = \frac{1}{\sqrt{ac-b^2}}\arctan\frac{au+b}{ac-b^2}
$$

于是

$$
\text{原式} = 2\int_{-\infty}^{+\infty} \frac{\mathrm{d}u}{au^2+2bu+c} = \begin{cases} \dfrac{2\pi}{\sqrt{ac-b^2}} & (a>0) \\ -\dfrac{2\pi}{\sqrt{ac-b^2}} & (a<0) \end{cases}
$$

3

$$
\begin{aligned}
&(1)\ \int_\Gamma xz^2\mathrm{d}x + yx^2\mathrm{d}y + zy^2\mathrm{d}z = \int_0^1 t^7\,\mathrm{d}t + t^4\,\mathrm{d}(t^2) + t^7\,\mathrm{d}(t^3) = \int_0^1 (3t^9 + t^7 + 2t^5)\,\mathrm{d}t = \frac{91}{120} \\
&(2)\ \int_\Gamma (y+z)\mathrm{d}x + (z+x)\mathrm{d}y + (x+y)\mathrm{d}z = \int_0^\pi \left(y^2 + yz + (x+z)(2z-2x) - xy - y^2\right)\,\mathrm{d}t \\
&\qquad = \int_0^\pi (2a+y)(z-x)\,\mathrm{d}t = \int_0^\pi (2a + a\sin 2t)a\cos 2t\,\mathrm{d}t = 0
\end{aligned}
$$

4

对 $\Gamma_1 : (1,0,0) \to (0,1,0)$，$x^2+y^2=1, z=0$，令 $\varphi(t) = (\cos t, \sin t, 0)$，于是

$$
\int_{\Gamma_1} (y^2-z^2)\,\mathrm{d}x + (z^2-x^2)\,\mathrm{d}y + (x^2-y^2)\,\mathrm{d}z = \int_0^{\frac{\pi}{2}} (-\sin^3 t - \cos^3 t)\,\mathrm{d}t = -\frac{4}{3}
$$

同理，对 $\Gamma_2 : (0,1,0) \to (0,0,1), \Gamma_3 : (0,0,1) \to (1,0,0)$，

$$
\int_{\Gamma_2} (y^2-z^2)\,\mathrm{d}x + (z^2-x^2)\,\mathrm{d}y + (x^2-y^2)\,\mathrm{d}z = -\frac{4}{3}, \int_{\Gamma_3} (y^2-z^2)\,\mathrm{d}x + (z^2-x^2)\,\mathrm{d}y + (x^2-y^2)\,\mathrm{d}z = -\frac{4}{3}
$$

于是

$$
\int_\Gamma (y^2-z^2)\,\mathrm{d}x + (z^2-x^2)\,\mathrm{d}y + (x^2-y^2)\,\mathrm{d}z = -4
$$

5

s 为 Γ 的弧长参数，于是

$$
\frac{\mathrm{d}\boldsymbol{p}}{\mathrm{d}s} = \boldsymbol{t}(s)
$$

是 Γ 在 s 处的单位切向量，则 $\mathrm{d}\boldsymbol{p} = \boldsymbol{t}(s)\mathrm{d}s$，于是

$$
\int_\Gamma \boldsymbol{F}\cdot\mathrm{d}\boldsymbol{p} = \int_\Gamma \boldsymbol{F}\cdot\boldsymbol{t}(s)\,\mathrm{d}s
$$

于是

$$
\left|\int_\Gamma \boldsymbol{F}\cdot\mathrm{d}\boldsymbol{p}\right| = \left|\int_\Gamma \boldsymbol{F}\cdot\boldsymbol{t}(s)\,\mathrm{d}s\right| \le \int_\Gamma |F\cdot\boldsymbol{t}(s)|\,\mathrm{d}s
$$

由 Cauchy-Schwarz 不等式

$$|F\cdot\boldsymbol{t}(s)|\le\|\boldsymbol{F}\|\cdot\|\boldsymbol{t}(s)\|\xlongequal{\boldsymbol{t}\text{为单位切向量}}\|\boldsymbol{F}\|$$

于是

$$\left|\int_\Gamma\boldsymbol{F}\cdot\mathrm{d}\boldsymbol{p}\right|\le\int_\Gamma|\boldsymbol{F}\cdot\boldsymbol{t}(s)|\,\mathrm{d}s\le\int_\Gamma\|\boldsymbol{F}\|\,\mathrm{d}s$$

6

(1) Σ 在 (x,y,z) 处的单位（内）法向量为 $\vec{n}=-(\frac{x}{a},\frac{y}{a},\frac{z}{a})$，于是

$$\iint_\Sigma x^4\mathrm{d}y\,\mathrm{d}z+y^4\mathrm{d}z\,\mathrm{d}x+z^4\mathrm{d}x\,\mathrm{d}y=\iint_\Sigma\langle\vec{v},\vec{n}\rangle\mathrm{d}\sigma=-\frac{1}{a}\iint_{x^2+y^2+z^2=a^2}x^5+y^5+z^5\,\mathrm{d}\sigma$$

由对称性可知

$$\iint_{x^2+y^2+z^2=a^2}x^5+y^5+z^5\,\mathrm{d}\sigma=0$$

于是

$$\iint_\Sigma x^4\mathrm{d}y\,\mathrm{d}z+y^4\mathrm{d}z\,\mathrm{d}x+z^4\mathrm{d}x\,\mathrm{d}y=0$$

(2) Σ 在 (x,y,z) 处的单位（外）法向量为 $\vec{n}=(\frac{x}{a},\frac{y}{a},\frac{z}{a})$，于是

$$\begin{aligned}\iint_\Sigma xz\,\mathrm{d}y\,\mathrm{d}z+yz\,\mathrm{d}z\,\mathrm{d}x+x^2\mathrm{d}x\,\mathrm{d}y&=\iint_\Sigma\langle\vec{v},\vec{n}\rangle\mathrm{d}\sigma=\frac{1}{a}\iint_{x^2+y^2+z^2=a^2}x^2z+y^2z+x^2z\\&=\frac{1}{a}\iint_{x^2+y^2+z^2=a^2}(2x^2+y^2)z\,\mathrm{d}\sigma\xlongequal{\text{对称性}}0\end{aligned}$$

(3) 由 Stokes 定理，设 $D=[0,a]\times[0,b]\times[0,c]$

$$\begin{aligned}\iint_\Sigma f(x)\,\mathrm{d}y\,\mathrm{d}z+g(y)\,\mathrm{d}z\,\mathrm{d}x+h(z)\,\mathrm{d}x\,\mathrm{d}y&=\iint_{\partial D}f(x)\,\mathrm{d}y\,\mathrm{d}z+g(y)\,\mathrm{d}z\,\mathrm{d}x+h(z)\,\mathrm{d}x\,\mathrm{d}y\\&=\iiint_D\left(\frac{\partial f(x)}{\partial x}+\frac{\partial g(y)}{\partial y}+\frac{\partial h(z)}{\partial z}\right)\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z\\&=\iiint_{[0,a]\times[0,b]\times[0,c]}\left(\frac{\partial f(x)}{\partial x}+\frac{\partial g(y)}{\partial y}+\frac{\partial h(z)}{\partial z}\right)\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z\\&=bc\int_0^a\frac{\partial f(x)}{\partial x}\,\mathrm{d}x+ac\int_0^b\frac{\partial g(y)}{\partial y}\,\mathrm{d}y+ab\int_0^c\frac{\partial h(z)}{\partial z}\,\mathrm{d}z\\&=bc(f(a)-f(0))+ac(g(b)-g(0))+ab(h(c)-h(0))\end{aligned}$$

(4) 考虑椭球的参数表示

$$\begin{gathered}x=a\sin\theta\cos\varphi,y=b\sin\theta\sin\varphi,z=c\cos\theta\\\frac{\partial(x,y)}{\partial(\theta,\varphi)}=\begin{vmatrix}a\cos\theta\cos\varphi&-a\sin\theta\sin\varphi\\b\cos\theta\sin\varphi&b\sin\theta\sin\varphi\end{vmatrix}=ab\cos\theta\sin\theta\end{gathered}$$

于是

$$\iint_\Sigma z\,\mathrm{d}x\,\mathrm{d}y=\iint_{[0,\pi]\times[0,2\pi]}abc\cos^2\theta\sin\theta=2\pi abc\int_0^\pi\cos^2\theta\sin\theta\,\mathrm{d}\theta=\frac{4\pi}{3}abc$$

(5) 设

$$\varphi(\theta,z)=(z\cos\theta,z\sin\theta,z),0\le z\le h,0\le\theta\le2\pi$$

$$\varphi^*\omega=\begin{vmatrix} z\sin\theta-z & z-z\cos\theta & z(\cos\theta-\sin\theta) \\ -z\sin\theta & z\cos\theta & 0 \\ \cos\theta & \sin\theta & 1 \end{vmatrix}=-z^2(\cos\theta-\sin\theta)$$

于是

$$\iint_\Sigma \omega=\iint_{[0,h]\times[0,2\pi]} -z^2(\cos\theta-\sin\theta)\,\mathrm{d}z\,\mathrm{d}\theta=-\int_0^h z^2\,\mathrm{d}z\int_0^{2\pi}(\cos\theta-\sin\theta)\,\mathrm{d}\theta=0$$

7

先考虑曲面 $\Sigma_1=\{(x,y,z)\mid x^2+y^2=R^2,0\le z\le h\}$，其定向由外法向量给出，$(x,y,z)$ 处单位外法向量为 $\vec{n}=(\frac{x}{R},\frac{y}{R},0)$，于是

$$\iint_{\Sigma_1}\boldsymbol{F}\,\mathrm{d}\boldsymbol{\sigma}=\iint_{\Sigma_1}\langle\boldsymbol{F},\vec{n}\rangle\,\mathrm{d}\sigma=\frac{1}{R}\iint_{\Sigma_1}(x+z)y\,\mathrm{d}\sigma\overset{\text{对称性}}{=\!=}0$$

再考虑曲面 $\Sigma_2=\{(x,y,z)\mid x^2+y^2\le R^2,z=0\}$，其定向由外法向量给出，$(x,y,z)$ 处单位外法向量为 $\vec{n}=(0,0,-1)$，于是

$$\iint_{\Sigma_2}\boldsymbol{F}\,\mathrm{d}\boldsymbol{\sigma}=\iint_{\Sigma_2}\langle\boldsymbol{F},\vec{n}\rangle\,\mathrm{d}\sigma=-\iint_{\Sigma_2}x\,\mathrm{d}\sigma\overset{\text{对称性}}{=\!=}0$$

对曲面 $\Sigma_2=\{(x,y,z)\mid x^2+y^2\le R^2,z=h\}$，其定向由外法向量给出，那么 (x,y,z) 处单位外法向量为 $\vec{n}=(0,0,1)$，于是

$$\iint_{\Sigma_3}\boldsymbol{F}\,\mathrm{d}\boldsymbol{\sigma}=\iint_{\Sigma_3}\langle\boldsymbol{F},\vec{n}\rangle\,\mathrm{d}\sigma=\iint_{\Sigma_3}x\,\mathrm{d}\sigma\overset{\text{对称性}}{=\!=}0$$

综上，

$$\iint_\Sigma\boldsymbol{F}\,\mathrm{d}\boldsymbol{\sigma}=\iint_{\Sigma_1}\boldsymbol{F}\,\mathrm{d}\boldsymbol{\sigma}+\iint_{\Sigma_2}\boldsymbol{F}\,\mathrm{d}\boldsymbol{\sigma}+\iint_{\Sigma_3}\boldsymbol{F}\,\mathrm{d}\boldsymbol{\sigma}=0$$

8

由课本 104 页 (10)，有

$$\iint_\Sigma P\,\mathrm{d}y\,\mathrm{d}z+Q\,\mathrm{d}z\,\mathrm{d}x+R\,\mathrm{d}x\,\mathrm{d}y=\pm\iint_D\left(-P\frac{\partial f}{\partial x}-Q\frac{\partial f}{\partial y}+R\right)\mathrm{d}x\,\mathrm{d}y$$

又因为 Σ 的法向量向上，与 z 轴夹角为锐角，故右边取正值，即

$$\iint_\Sigma P\,\mathrm{d}y\,\mathrm{d}z+Q\,\mathrm{d}z\,\mathrm{d}x+R\,\mathrm{d}x\,\mathrm{d}y=\iint_D\left(-P\frac{\partial f}{\partial x}-Q\frac{\partial f}{\partial y}+R\right)\mathrm{d}x\,\mathrm{d}y$$

(1) 上式取 $Q=R=0$ 即可得证

(2) 上式取 $P=R=0$ 即可得证

9

(1) 记 $S=\{(x,y)\mid\frac{x^2}{a^2}+\frac{y^2}{b^2}\le1\}$，于是

$$\text{原式}=\int_{\partial S}(x+y)\,\mathrm{d}x-(x-y)\,\mathrm{d}y=\iint_S(-1-1)\,\mathrm{d}x\wedge\mathrm{d}y=-2\iint_S\mathrm{d}x\,\mathrm{d}y=-2\pi ab$$

(2) 设 $S=\{(x,y)\mid(x-\frac{a}{2})^2+y^2\le\frac{a^2}{4},y\ge0\}$，$\Gamma_1=\{(x,0)\mid0\le x\le a\}$。于是 $\partial S=\Gamma\cup\Gamma_1$，

$$\int_{\partial S} e^x \sin y \,\mathrm{d}x + e^x \cos y \,\mathrm{d}y = \int_S (e^x \cos y - e^x \cos y) \,\mathrm{d}x \,\mathrm{d}y = 0$$

$$\int_{\Gamma_1} e^x \sin y \,\mathrm{d}x + e^x \cos y \,\mathrm{d}y = \int_{\Gamma_1} e^x \sin 0 \,\mathrm{d}x = 0$$

于是

$$\int_{\Gamma_1} e^x \sin y \,\mathrm{d}x + e^x \cos y \,\mathrm{d}y = \int_{\partial S} e^x \sin y \,\mathrm{d}x + e^x \cos y \,\mathrm{d}y - \int_{\Gamma_1} e^x \sin y \,\mathrm{d}x + e^x \cos y \,\mathrm{d}y = 0$$

10

(1) 令 $y = x \tan\theta$，于是 $\theta \in [-\frac{\pi}{4}, \frac{\pi}{4}]$，双扭线化为

$$\left(\frac{x^2}{\cos^2\theta}\right)^2 = a^2 x^2 \frac{\cos 2\theta}{\cos^2\theta} \implies x^2 = a^2 \cos 2\theta \cos^2\theta$$

设该双扭线围成的区域是 S，记

$$V = \{(x, \theta) \mid x^2 = a^2 \cos 2\theta \cos^2\theta, \theta \in \left[-\frac{\pi}{4}, \frac{\pi}{4}\right]\}$$

$$\begin{aligned}\int_S \mathrm{d}x \,\mathrm{d}y &= \frac{1}{2}\int_{\partial S} x \,\mathrm{d}y - y \,\mathrm{d}x = \frac{1}{2}\int_V (x\tan\theta - x\tan\theta)\,\mathrm{d}x + \frac{x^2}{\cos^2\theta}\,\mathrm{d}\theta = \frac{1}{2}\int_V \frac{a^2\cos 2\theta\cos^2\theta}{\cos^2\theta}\,\mathrm{d}\theta \\ &= \frac{a^2}{2}\cdot 2\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \cos 2\theta \,\mathrm{d}\theta = a^2\end{aligned}$$

(2) 令 $y = xt$，于是 $t \in [0, +\infty)$，Descartes 叶形线化为

$$x^3(1+t^3) = 3atx^2 \implies x = \frac{3at}{1+t^3}$$

设该Descartes 叶形线围成的区域是 S，记

$$V = \{(x, t) \mid x = \frac{3at}{1+t^2}, t \in [0, +\infty)\}$$

于是

$$\int_S \mathrm{d}x \,\mathrm{d}y = \frac{1}{2}\int_{\partial S} x \,\mathrm{d}y - y \,\mathrm{d}x = \frac{1}{2}\int_V (tx - tx)\,\mathrm{d}x + x^2 \,\mathrm{d}t = \frac{1}{2}\int_V x^2 \,\mathrm{d}t = \frac{9a^2}{2}\int_0^{+\infty} \frac{t^2}{(t^3+1)^2}\,\mathrm{d}t$$

其中

$$\int \frac{t^2}{(t^3+1)^2}\,\mathrm{d}t \overset{u=t^3}{=\!=\!=} \frac{1}{3}\int \frac{\mathrm{d}u}{(u+1)^2} = -\frac{1}{3}\frac{1}{u+1} + C$$

于是

$$\int_S \mathrm{d}x \,\mathrm{d}y = \frac{9a^2}{2}\int_0^{+\infty} \frac{t^2}{(t^3+1)^2}\,\mathrm{d}t = \frac{9a^2}{2}\cdot\frac{1}{3} = \frac{3a^2}{2}$$

11

$$A = \frac{1}{2}\int_\Gamma x \,\mathrm{d}y - y \,\mathrm{d}x = \frac{1}{2}\int_\alpha^\beta (\varphi(t)\psi'(t) - \psi(t)\varphi'(t))\,\mathrm{d}t = \frac{1}{2}\int_\alpha^\beta \begin{vmatrix}\varphi(t) & \psi(t) \\ \varphi'(t) & \psi'(t)\end{vmatrix}\,\mathrm{d}t$$

12

设 S 是由 Γ 围成的区域，于是

$$\int_\Gamma f(x^n+y^n)(x^{n-1}\,\mathrm{d}x+y^{n-1}\,\mathrm{d}y)=\int_S -x^{n-1}y^{n-1}f'(x^n+y^n)+x^{n-1}y^{n-1}f'(x^n+y^n)\,\mathrm{d}x\,\mathrm{d}y=0$$

13

由教材 80 页式 (2)，我们有第一型曲线积分表达的 Green 公式

$$\int_\Gamma (P\cos(\boldsymbol{n},\boldsymbol{i})+Q\cos(\boldsymbol{n},\boldsymbol{i}))\,\mathrm{d}s=\iint_D\left(\frac{\partial P}{\partial x}+\frac{\partial Q}{\partial y}\right)\mathrm{d}x\,\mathrm{d}y$$

设 $\boldsymbol{\alpha}=(a_1,a_2)$，其中 a_1,a_2 为常数。因为 $\boldsymbol{\alpha},\boldsymbol{n}$ 都是单位向量于是

$$\cos(\boldsymbol{\alpha},\boldsymbol{n})=\frac{\langle\boldsymbol{\alpha},\boldsymbol{n}\rangle}{\|\boldsymbol{\alpha}\|\cdot\|\boldsymbol{n}\|}=\langle\boldsymbol{\alpha},\boldsymbol{n}\rangle=\langle a_1\boldsymbol{i},\boldsymbol{n}\rangle+\langle a_2\boldsymbol{j},\boldsymbol{n}\rangle=a_1\cos(\boldsymbol{n},\boldsymbol{i})+a_2\cos(\boldsymbol{n},\boldsymbol{j})$$

于是

$$\int_\Gamma\cos(\boldsymbol{\alpha},\boldsymbol{n})\,\mathrm{d}s=\int_\Gamma(a_1\cos(\boldsymbol{n},\boldsymbol{i})+a_2\cos(\boldsymbol{n},\boldsymbol{j}))\,\mathrm{d}s=\iint_D\left(\frac{\partial a_1}{\partial x}+\frac{\partial a_2}{\partial y}\right)\mathrm{d}x\,\mathrm{d}y=\iint_D 0\,\mathrm{d}x\,\mathrm{d}y=0$$

14

由第一型曲线积分表达的 Green 公式

$$\int_\Gamma(x\cos(\boldsymbol{n},\boldsymbol{i})+y\cos(\boldsymbol{n},\boldsymbol{j}))\,\mathrm{d}s=\iint_D\left(\frac{\partial x}{\partial x}+\frac{\partial y}{\partial y}\right)\mathrm{d}x\,\mathrm{d}y=2\iint_D\mathrm{d}x\,\mathrm{d}y=2V(D)$$

其中 D 为 Γ 围成的区域，$V(D)$ 为其面积。

15

设 $\omega=\dfrac{(x+y)\,\mathrm{d}x-(x-y)\,\mathrm{d}y}{x^2+y^2}$，于是

$$\mathrm{d}\omega=\left(-\frac{x^2-2xy-y^2}{(x^2+y^2)^2}-\frac{-x^2+2xy+y^2}{(x^2+y^2)^2}\right)\mathrm{d}x\wedge\mathrm{d}y=0$$

设 D 为 L 与 x 轴围成的区域。记 $B_+(0,r)=\{(x,y)\mid x^2+y^2<r^2,y\geq 0\}$ 即 $B(0,r)$ 在上半平面中的部分，记 $S_+(0,r)=\{(x,y)\mid x^2+y^2=r^2,y\geq 0\}$。

取充分小的 ε 使得 $B_+(0,2\varepsilon)\subset D$，注意到在 $L_1=\{(x,0)\mid x\in[-1,-\varepsilon]\cup[\varepsilon,1]\}$（方向沿 x 递减的方向）上有

$$\int_{L_1}\frac{(x+y)\,\mathrm{d}x-(x-y)\,\mathrm{d}y}{x^2+y^2}=\int_{L_1}\frac{\mathrm{d}x}{x}=0$$

于是

$$\begin{aligned}\int_L\frac{(x+y)\,\mathrm{d}x-(x-y)\,\mathrm{d}y}{x^2+y^2}&=\int_{S_+(0,\varepsilon)}\frac{(x+y)\,\mathrm{d}x-(x-y)\,\mathrm{d}y}{x^2+y^2}=\frac{1}{\varepsilon^2}\int_{S_+(0,\varepsilon)}(x+y)\,\mathrm{d}x-(x-y)\,\mathrm{d}y\\&=\frac{1}{\varepsilon^2}\int_\pi^0\varepsilon^2(-\sin^2\theta-\cos^2\theta)\,\mathrm{d}\theta=\pi\end{aligned}$$

16

设 $\omega=\dfrac{e^x}{x^2+y^2}((x\sin y-y\cos y)\,\mathrm{d}x+(x\cos y+y\sin y)\,\mathrm{d}y)$，则

$$\begin{aligned}\mathrm{d}\omega =&\left(\frac{2ye^x\left(x\sin y-y\cos y\right)}{(x^2+y^2)^2}-\frac{e^x\left(x\cos y-\cos y+y\sin y\right)}{x^2+y^2}\right.\\&\left.+\frac{e^x(x^2+y^2)-2xe^x}{(x^2+y^2)^2}(x\cos y+y\sin y)+\frac{e^x\cos y}{(x^2+y^2)}\right)\mathrm{d}x\wedge\mathrm{d}y\\=&\left(-\frac{e^x\left(2y^2\cos y+2x^2\cos y\right)}{(x^2+y^2)^2}+\frac{2e^x\cos y}{(x^2+y^2)}\right)\mathrm{d}x\wedge\mathrm{d}y=0\end{aligned}$$

于是，取充分小的 ε 使得 $B(0,2\varepsilon)\subset D$，则

$$\int_\Gamma\omega=\int_{\partial B(0,\varepsilon)}\omega\overset{x=\varepsilon\cos\theta,y=\varepsilon\sin\theta}{=\!=\!=}\int_0^{2\pi}e^{\varepsilon\cos\theta}\cos(\varepsilon\sin\theta)\,\mathrm{d}\theta$$

现在我们只需计算

$$\int_0^{2\pi}e^{\varepsilon\cos\theta}\cos(\varepsilon\sin\theta)\,\mathrm{d}\theta$$

记

$$f(t)=\int_0^{2\pi}e^{t\cos\theta}\cos(t\sin\theta)\,\mathrm{d}\theta$$

则

$$f'(t)=\int_0^{2\pi}\cos\theta e^{t\cos\theta}\cos(t\sin\theta)-\sin\theta e^{t\cos\theta}\sin(t\sin\theta)\,\mathrm{d}\theta$$

其中

$$\int\cos\theta e^{t\cos\theta}\cos(t\sin\theta)-\sin\theta e^{t\cos\theta}\sin(t\sin\theta)\,\mathrm{d}\theta=\frac{1}{t}\sin(t\sin\theta)e^{t\cos\theta}+C$$

于是

$$f'(t)=\int_0^{2\pi}\cos\theta e^{t\cos\theta}\cos(t\sin\theta)-\sin\theta e^{t\cos\theta}\sin(t\sin\theta)\,\mathrm{d}\theta=\left[\frac{1}{t}\sin(t\sin\theta)e^{t\cos\theta}\right]\Bigg|_0^{2\pi}=0$$

则 $f(t)=f(0)=2\pi,\forall t\in\mathbb{R}$，于是

$$\int_\Gamma\omega=\int_0^{2\pi}e^{\varepsilon\cos\theta}\cos(\varepsilon\sin\theta)\,\mathrm{d}\theta=2\pi$$

17

$$\frac{\partial u}{\partial\boldsymbol{n}}=\cos(\boldsymbol{n},\boldsymbol{i})\frac{\partial u}{\partial x}+\cos(\boldsymbol{n},\boldsymbol{j})\frac{\partial u}{\partial y}$$

于是由第一型曲线积分表达的 Green 公式

$$\begin{aligned}\int_\Gamma\begin{vmatrix}\frac{\partial u}{\partial\boldsymbol{n}}&\frac{\partial v}{\partial\boldsymbol{n}}\\u&v\end{vmatrix}\mathrm{d}s&=\int_\Gamma\left(\left(v\frac{\partial u}{\partial x}-u\frac{\partial v}{\partial x}\right)\cos(\boldsymbol{n},\boldsymbol{i})+\left(v\frac{\partial u}{\partial y}-u\frac{\partial v}{\partial y}\right)\cos(\boldsymbol{n},\boldsymbol{j})\right)\\&=\iint_G\left(\left(\frac{\partial v}{\partial x}\frac{\partial u}{\partial x}+v\frac{\partial^2u}{\partial x^2}-\frac{\partial u}{\partial x}\frac{\partial v}{\partial x}-u\frac{\partial^2v}{\partial x^2}\right)+\left(\frac{\partial v}{\partial y}\frac{\partial u}{\partial y}+v\frac{\partial^2u}{\partial y^2}-\frac{\partial u}{\partial y}\frac{\partial v}{\partial y}-u\frac{\partial^2v}{\partial y^2}\right)\right)\mathrm{d}x\,\mathrm{d}y\\&=\iint_G(v\Delta u-u\Delta v)\,\mathrm{d}x\,\mathrm{d}y=\iint_G\begin{vmatrix}\Delta u&\Delta v\\u&v\end{vmatrix}\mathrm{d}x\,\mathrm{d}y\end{aligned}$$

取足够小的 ε 使得 $B((x_0,y_0),2\varepsilon)\subset G$。

在本题中，取 $v=\ln r$，于是在 $G\setminus B((x_0,y_0),\varepsilon)$ 上

$$\frac{\partial \ln v}{\partial x}=\frac{x-x_0}{(x-x_0)^2+(y-y_0)^2},\ \frac{\partial^2 \ln v}{\partial x^2}=\frac{(y-y_0)^2-(x-x_0)^2}{((x-x_0)^2+(y-y_0)^2)^2}$$

同理

$$\frac{\partial^2 \ln v}{\partial y^2}=\frac{(x-x_0)^2-(y-y_0)^2}{((x-x_0)^2+(y-y_0)^2)^2}$$

故 $\Delta \ln v=\frac{\partial^2 \ln v}{\partial x^2}+\frac{\partial^2 \ln v}{\partial y^2}=0$。于是

$$\begin{aligned}\int_\Gamma\left(u\frac{\partial \ln r}{\partial n}-\ln r\frac{\partial u}{\partial n}\right)\mathrm{d}s&=\iint_{G\setminus B((x_0,y_0),\varepsilon)}\begin{vmatrix}\Delta u & \Delta v\\ u & v\end{vmatrix}\mathrm{d}x\,\mathrm{d}y+\int_{\partial B((x_0,y_0),\varepsilon}\left(u\frac{\partial \ln r}{\partial n}-\ln r\frac{\partial u}{\partial n}\right)\mathrm{d}s\\&=\int_{\partial B((x_0,y_0),\varepsilon}\left(u\frac{\partial \ln r}{\partial n}-\ln r\frac{\partial u}{\partial n}\right)\mathrm{d}s\\&=\int_{\partial B((x_0,y_0),\varepsilon}u\frac{\partial \ln r}{\partial n}\,\mathrm{d}s-\ln\varepsilon\int_{\partial B((x_0,y_0),\varepsilon}\left(\cos(\boldsymbol{n},\boldsymbol{i})\frac{\partial u}{\partial x}+\cos(\boldsymbol{n},\boldsymbol{j})\frac{\partial u}{\partial y}\right)\mathrm{d}s\\&=\int_{\partial B((x_0,y_0),\varepsilon}u\frac{\partial \ln r}{\partial n}\,\mathrm{d}s-\ln\varepsilon\iint_{B((x_0,y_0),\varepsilon)}\Delta u\,\mathrm{d}s\\&=\int_{\partial B((x_0,y_0),\varepsilon}u\frac{\partial \ln r}{\partial n}\,\mathrm{d}s\end{aligned}$$

注意到这里 $\boldsymbol{n}=(\frac{x-x_0}{\varepsilon},\frac{y-y_0}{\varepsilon})$，于是

$$\frac{\partial \ln r}{\partial n}=\frac{1}{\varepsilon}\left((x-x_0)\frac{\partial \ln r}{\partial x}+(y-y_0)\frac{\partial \ln r}{\partial y}\right)=\frac{1}{\varepsilon}$$

于是

$$\int_\Gamma\left(u\frac{\partial \ln r}{\partial n}-\ln r\frac{\partial u}{\partial n}\right)\mathrm{d}s=\frac{1}{\varepsilon}\int_{\partial B((x_0,y_0),\varepsilon}u\,\mathrm{d}s$$

由积分中值定理

$$\int_\Gamma\left(u\frac{\partial \ln r}{\partial n}-\ln r\frac{\partial u}{\partial n}\right)\mathrm{d}s=\frac{1}{\varepsilon}\int_{\partial B((x_0,y_0),\varepsilon}u\,\mathrm{d}s=\frac{1}{\varepsilon}\cdot\varepsilon 2\pi u(x_1,y_1)$$

由 u 的连续性，当 $\varepsilon\to 0$ 时

$$\int_\Gamma\left(u\frac{\partial \ln r}{\partial n}-\ln r\frac{\partial u}{\partial n}\right)\mathrm{d}s=\frac{1}{\varepsilon}\cdot\varepsilon 2\pi u(x_1,y_1)=2\pi u(x_0,y_0)$$

于是

$$u(x_0,y_0)=\frac{1}{2\pi}\int_\Gamma\left(u\frac{\partial \ln r}{\partial n}-\ln r\frac{\partial u}{\partial n}\right)\mathrm{d}s$$

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由上一题

$$\frac{\partial \ln r}{\partial n}=\frac{1}{R}$$

$$\begin{aligned}u(x_0,y_0)=\frac{1}{2\pi}\int_\Gamma\left(u\frac{\partial \ln r}{\partial n}-\ln r\frac{\partial u}{\partial n}\right)\mathrm{d}s&=\frac{1}{2\pi R}\int_\Gamma u(x,y)\,\mathrm{d}s-\frac{\ln R}{2\pi}\int_\Gamma\frac{\partial u}{\partial n}\,\mathrm{d}s\\&=\frac{1}{2\pi R}\int_\Gamma u(x,y)\,\mathrm{d}s-\frac{\ln R}{2\pi}\iint_G\Delta u\,\mathrm{d}x\,\mathrm{d}y=\frac{1}{2\pi R}\int_\Gamma u(x,y)\,\mathrm{d}s\end{aligned}$$

19

(1) 记 $S=\{(x,y,z)\mid x,y,z\geq 0,x+y+z\leq 1\}$，于是

$$\begin{aligned}\text{原式}&=\iint_{\partial S}xy\,\mathrm{d}y\,\mathrm{d}z+yz\,\mathrm{d}z\,\mathrm{d}x+zx\,\mathrm{d}x\,\mathrm{d}y=\iiint_S(y+z+x)\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z=3\iiint_S x\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z\\&=3\int_0^1\mathrm{d}x\int_0^{1-x}\mathrm{d}y\int_0^{1-x-y}x\,\mathrm{d}z=3\int_0^1\mathrm{d}x\int_0^{1-x}(1-x-y)x\,\mathrm{d}y=3\int_0^1\frac{1}{2}x(1-x)^2\,\mathrm{d}x=\frac{1}{8}\end{aligned}$$

(2) 记 $S=\{(x,y,z)\mid z^2=x^2+y^2,0\leq z\leq 1\}$，于是

$$\partial S=\Sigma\cup\{(x,y,z)\mid z=1,x^2+y^2\leq 1\}$$

记 $\Sigma_1=\{(x,y,z)\mid z=1,x^2+y^2\leq 1\}$，于是

$$\begin{aligned}\iint_{\partial S}x^2\,\mathrm{d}y\,\mathrm{d}z+y^2\,\mathrm{d}z\,\mathrm{d}x+z^2\,\mathrm{d}x\,\mathrm{d}y&=2\iiint_S(x+y+z)\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z=2\int_0^1\mathrm{d}z\int_0^z\mathrm{d}r\int_0^{2\pi}r(z+r\sin\theta+\mathrm{r}\cos\theta)\,\mathrm{d}\theta\\&=4\pi\int_0^1\mathrm{d}z\int_0^z rz\,\mathrm{d}r=2\pi\int_0^1 z^3\,\mathrm{d}z=\frac{\pi}{2}\\\iint_{\Sigma_1}x^2\,\mathrm{d}y\,\mathrm{d}z+y^2\,\mathrm{d}z\,\mathrm{d}x+z^2\,\mathrm{d}x\,\mathrm{d}y&=\iint_{\Sigma_1}\mathrm{d}x\,\mathrm{d}y=\pi\end{aligned}$$

于是

$$\text{原式}=\iint_{\partial S}x^2\,\mathrm{d}y\,\mathrm{d}z+y^2\,\mathrm{d}z\,\mathrm{d}x+z^2\,\mathrm{d}x\,\mathrm{d}y-\iint_{\Sigma_1}x^2\,\mathrm{d}y\,\mathrm{d}z+y^2\,\mathrm{d}z\,\mathrm{d}x+z^2\,\mathrm{d}x\,\mathrm{d}y=-\frac{\pi}{2}$$

20

设 $\boldsymbol{e}=(e_1,e_2,e_3)$，因为 $\boldsymbol{e}$ 为一个固定的向量，则 e_1,e_2,e_3 为常数，于是

$$\cos(\boldsymbol{e},\boldsymbol{n})=e_1\langle\boldsymbol{n},\boldsymbol{i}\rangle+e_2\langle\boldsymbol{n},\boldsymbol{j}\rangle+e_3\langle\boldsymbol{n},\boldsymbol{k}\rangle=e_1\cos(\boldsymbol{n},\boldsymbol{i})+e_2\cos(\boldsymbol{n},\boldsymbol{j})+e_3\cos(\boldsymbol{n},\boldsymbol{k})$$

于是，使用第一型曲面积分表达的 Gauss 公式

$$\int_{\partial\Omega}\cos(\boldsymbol{e},\boldsymbol{n})\,\mathrm{d}\sigma=\int_{\partial\Omega}(e_1\cos(\boldsymbol{n},\boldsymbol{i})+e_2\cos(\boldsymbol{n},\boldsymbol{j})+e_3\cos(\boldsymbol{n},\boldsymbol{k}))\,\mathrm{d}\sigma=\int_\Omega\left(\frac{\partial e_1}{\partial x}+\frac{\partial e_2}{\partial y}+\frac{\partial e_3}{\partial z}\right)\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z=0$$

21

$$\cos(\boldsymbol{p},\boldsymbol{n})=\frac{\langle\boldsymbol{p},\boldsymbol{n}\rangle}{\|\boldsymbol{p}\|\|\boldsymbol{n}\|}=\frac{1}{p}((x-a)\cos(\boldsymbol{n},\boldsymbol{i})+(y-b)\cos(\boldsymbol{n},\boldsymbol{j})+(z-c)\cos(\boldsymbol{n},\boldsymbol{k}))$$

于是

$$\begin{aligned}\int_{\partial\Omega}\cos(\boldsymbol{p},\boldsymbol{n})\,\mathrm{d}\sigma&=\int_{\partial\Omega}\frac{1}{p}((x-a)\cos(\boldsymbol{n},\boldsymbol{i})+(y-b)\cos(\boldsymbol{n},\boldsymbol{j})+(z-c)\cos(\boldsymbol{n},\boldsymbol{k}))\\&=\iiint_\Omega\left(\frac{\partial\frac{x-a}{p}}{\partial x}+\frac{\partial\frac{y-b}{p}}{\partial x}+\frac{\partial\frac{z-c}{p}}{\partial x}\right)\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z\\&=\iiint_\Omega\left(\frac{1}{p}-\frac{(x-a)^2}{p^3}+\frac{1}{p}-\frac{(y-b)^2}{p^3}+\frac{1}{p}-\frac{(z-c)^2}{p^3}\right)\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z\\&=\iiint_\Omega\frac{2}{p}\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z\end{aligned}$$

即

$$\frac{1}{2}\int_{\partial\Omega}\cos(\boldsymbol{p},\boldsymbol{n})\,\mathrm{d}\sigma=\iiint_\Omega\frac{\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z}{p}$$

22

(1) 记 $S=\{(x,y,z)\mid x^2+y^2+z^2\le a^2,x+y+z=0\}$，其定向由单位法向量 $\boldsymbol{n}=(\frac{\sqrt3}{3},\frac{\sqrt3}{3},\frac{\sqrt3}{3})$ 给出，于是

$$\begin{aligned}\int_\Gamma y\,\mathrm{d}x+z\,\mathrm{d}y+x\,\mathrm{d}z&=\iint_S(0-1)\,\mathrm{d}y\,\mathrm{d}z+(0-1)\,\mathrm{d}z\,\mathrm{d}x+(0-1)\,\mathrm{d}x\,\mathrm{d}y=-\iint_S\mathrm{d}y\,\mathrm{d}z+\mathrm{d}z\,\mathrm{d}x+\mathrm{d}x\,\mathrm{d}y\\&=-\iint_S\sqrt3\,\mathrm{d}\sigma=-\sqrt3\pi a^2\end{aligned}$$

(2) 记 $S=\{(x,y,z)\mid x^2+(y-1)^2\le1,y=z\}$，其定向由单位法向量 $\boldsymbol{n}=(0,1,-1)$ 给出，于是

$$\text{原式}=\iint_S0\,\mathrm{d}y\,\mathrm{d}z+0\,\mathrm{d}z\,\mathrm{d}x+0\,\mathrm{d}x\,\mathrm{d}y=0$$

(3) 不妨设 $a\ge0$，记 $S=\{(x,y,z)\mid x^2+y^2+z^2\le a^2,x+y+z=a\}$，其定向由单位法向量 $\boldsymbol{n}=(-\frac{\sqrt3}{3},-\frac{\sqrt3}{3},-\frac{\sqrt3}{3})$ 给出，于是

$$\int_\Gamma y^2\,\mathrm{d}x+z^2\mathrm{d}y+x^2\,\mathrm{d}z=-2\int_S z\,\mathrm{d}y\,\mathrm{d}z+x\,\mathrm{d}z\,\mathrm{d}x+y\,\mathrm{d}x\,\mathrm{d}y=\frac{2\sqrt3}{3}\int_S(x+y+z)\,\mathrm{d}\sigma=\frac{2a\sqrt3}{3}\int_S\mathrm{d}\sigma=\frac{4\sqrt3\pi a^3}{9}$$

23

设 $\boldsymbol{a}=(a_1,a_2,a_3),\boldsymbol{p}=(x,y,z)$，其中 a_1,a_2,a_3 为常数，于是

$$\boldsymbol{a}\times\boldsymbol{p}=(a_2z-a_3y,a_3x-a_1z,a_1y-a_2x)$$

于是

$$\begin{aligned}\int_{\partial\Sigma}\boldsymbol{a}\times\boldsymbol{p}\cdot\mathrm{d}\boldsymbol{p}&=\int_{\partial\Sigma}(a_2z-a_3y)\,\mathrm{d}x+(a_3x-a_1z)\,\mathrm{d}y+(a_1y-a_2x)\,\mathrm{d}z\\&=2\iint_\Sigma a_1\,\mathrm{d}y\,\mathrm{d}z+a_2\,\mathrm{d}z\,\mathrm{d}x+a_3\,\mathrm{d}x\,\mathrm{d}y\\&=2\iint_\Sigma\boldsymbol{a}\cdot\boldsymbol{n}\,\mathrm{d}\sigma\end{aligned}$$

24

设 $S=\{(x,y,z)\mid x^2+y^2+z^2\le4,x+y=2\}$，其定向由单位法向量 $\boldsymbol{n}=(\frac{\sqrt2}{2},\frac{\sqrt2}{2},0)$ 给出，于是

$$\int_\Gamma y\,\mathrm{d}x+z\,\mathrm{d}y+x\,\mathrm{d}z=-\iint_S\mathrm{d}y\,\mathrm{d}z+\mathrm{d}z\,\mathrm{d}x+\mathrm{d}x\,\mathrm{d}y=-\iint_S\sqrt2\,\mathrm{d}\sigma=-2\sqrt2\pi$$

25

设 $S=\{(x,y,z)\mid x^2+y^2\le1,z=xy\}$，于是

$$\int_\Gamma y\,\mathrm{d}x+z\,\mathrm{d}y+x\,\mathrm{d}z=-\iint_S\mathrm{d}y\,\mathrm{d}z+\mathrm{d}z\,\mathrm{d}x+\mathrm{d}x\,\mathrm{d}y=-\iint_S(-x-y+1)\,\mathrm{d}x\,\mathrm{d}y$$

由对称性

$$\iint_S(-x-y+1)\,\mathrm{d}x\,\mathrm{d}y=\iint_S\mathrm{d}x\,\mathrm{d}y=\pi$$

于是

$$\int_\Gamma y\,\mathrm{d}x+z\,\mathrm{d}y+x\,\mathrm{d}z=-\pi$$

26

设 $S=\{(x,y,z)\mid x^2+y^2\le ax,x^2+y^2+z^2=a^2\}$，于是所求的功为

$$\int_\Gamma y^2\,\mathrm{d}x+z^2\,\mathrm{d}y+x^2\,\mathrm{d}z=-2\int_S z\,\mathrm{d}y\,\mathrm{d}z+x\,\mathrm{d}z\,\mathrm{d}x+y\,\mathrm{d}x\,\mathrm{d}y$$

这里 $z=f(x,y)=\sqrt{a^2-x^2-y^2}$，于是

$$\frac{\partial f}{\partial x}=-\frac{x}{\sqrt{a^2-x^2-y^2}},\frac{\partial f}{\partial y}=-\frac{y}{\sqrt{a^2-x^2-y^2}}$$

于是

$$\begin{aligned}\int_S z\,\mathrm{d}y\,\mathrm{d}z+x\,\mathrm{d}z\,\mathrm{d}x+y\,\mathrm{d}x\,\mathrm{d}y&=\int_S\left(x+y+\frac{xy}{\sqrt{a^2-x^2-y^2}}\right)\mathrm{d}x\,\mathrm{d}y\\&\xlongequal{x-a/2=r\cos\theta,y=r\sin\theta}\int_{2\pi}^0\mathrm{d}\theta\int_0^{\frac{a}{2}}\left(\frac{a}{2}+r\cos\theta+r\sin\theta+\frac{r^2\sin\theta\cos\theta}{\sqrt{a^2-r^2}}\right)r\,\mathrm{d}r\\&=-\frac{a^3\pi}{8}\end{aligned}$$

$$\int_\Gamma y^2\,\mathrm{d}x+z^2\,\mathrm{d}y+x^2\,\mathrm{d}z=-2\int_S z\,\mathrm{d}y\,\mathrm{d}z+x\,\mathrm{d}z\,\mathrm{d}x+y\,\mathrm{d}x\,\mathrm{d}y=\frac{a^3\pi}{4}$$

27

$$\int_\Gamma\boldsymbol{F}\cdot\mathrm{d}\boldsymbol{p}=\int_\Gamma m\boldsymbol{a}\cdot\,\mathrm{d}\boldsymbol{p}=m\int_\Gamma\frac{\mathrm{d}\boldsymbol{v}}{\mathrm{d}t}\cdot\mathrm{d}\boldsymbol{p}=m\int_\Gamma\frac{\mathrm{d}\boldsymbol{p}}{\mathrm{d}t}\,\mathrm{d}\boldsymbol{v}=m\int_\Gamma\boldsymbol{v}\,\mathrm{d}\boldsymbol{v}=\frac{1}{2}m\boldsymbol{v}^2(\boldsymbol{p})\Big|_{\boldsymbol{a}}^{\boldsymbol{b}}$$